

NSCT (HEC) IT Graduate Preparation Guide

1. Programming Fundamentals

C/C++, variables, loops, conditions, functions, OOP concepts (Encapsulation, Inheritance, Polymorphism, Abstraction).

2. Data Structures & Algorithms

Arrays, Linked Lists, Stacks, Queues, Trees, Graphs, Sorting (Merge, Quick), Searching (Binary Search), Big-O complexity.

3. Database Systems

SQL (SELECT, JOIN), Normalization (1NF, 2NF, 3NF), ER diagrams, ACID properties.

4. Operating Systems

Processes vs Threads, Scheduling, Deadlocks (4 conditions), Memory management.

5. Computer Networks

OSI model (7 layers), TCP/IP, DNS, HTTP, IP addressing.

6. Software Engineering

SDLC, Agile vs Waterfall, Testing types, UML diagrams.

7. Web & Emerging Technologies

HTML, CSS, JavaScript basics, APIs, Cloud basics, AI basics.

8. Analytical & IQ

Logical reasoning, patterns, basic math, problem solving.

Sample MCQs

1. Binary Search complexity? $O(\log n)$ 2. OSI layers? 7 3. Primary key? Unique identifier 4. HTTP port? 80 5. Stack principle? LIFO